DATA-RIDE Policy Recommendation

Reflections from Citizen-Generated Data in Nancy

DATA-RIDE is a citizen science and advocacy project developed within the LAUDS Factories programme in collaboration with Université de Lorraine.

The project explores how citizen-generated data can contribute to public debate and support evidence-informed reflections on urban mobility. Between February and May 2026, ten volunteers used open-source senseBox devices to collect data on everyday cycling conditions in the Nancy area. The devices recorded GPS position, speed, lateral distance, particulate matter and motion-based indicators related to stopping and road-surface conditions.

The campaign generated 193,328 observations across 357 trips, covering more than 1,000 km and approximately 84 hours of valid recorded movement. The data were cleaned, processed and linked to OpenStreetMap street segments, together with selected local datasets on cycling infrastructure, parking and tree presence.

DATA-RIDE sought to better understand how cycling is experienced in practice: where cyclists slow down or stop, where overtaking distances are shorter, where road surfaces are rougher, how the cars interact with cyclists, and which conditions contribute to comfort or insecurity.

Data collection was followed by a participatory interpretation process. Results were presented and discussed during the Fête du Vélo and a workshop held at the Lorraine Fab Living Lab on 1 June, involving cyclists, researchers, local organisations and mobility stakeholders.

The findings presented in this document therefore combine three elements: measured sensor data with spatial processing, observations and experiences shared during the participatory activities, and broader policy recommendations. The maps presented in the accompanying presentation are also available on the DATA-RIDE project page on re:edu github https://github.com/reedu-reengineering-education/dataride-lauds. These maps help connect the findings to specific streets, corridors and urban conditions in Nancy, creating a connection in which evidence, experience and local knowledge could

be brought into dialogue.

Why Citizen-Generated Data Matters

During the DATA-RIDE discussions, cyclists mentioned that mobility authorities increasingly rely on digital data sources such as Strava and Geovelo to support planning and decision-making. Participants also noted that the value of DATA-RIDE lies not only in the measurements themselves, but in its capacity to reveal aspects of everyday cycling experience that are often difficult to capture through conventional planning tools.

At the same time, participants emphasised that the current dataset remains limited and should be understood as exploratory rather than representative of all cycling conditions in Nancy. The usefulness of the system would increase through larger-scale deployment and wider participation, similarly to community-based platforms such as Waze, where the collective contribution of many users generates increasingly robust insights.

Nevertheless, even though the current data collection is neither exhaustive nor comprehensive, the findings provide evidence supporting concerns that cyclists in Nancy have raised for many years. In particular, they suggest that the feeling of insecurity reported by many cyclists, even on streets where cycling infrastructure exists, is not merely a matter of perception.

Among the available indicators, the distance at which vehicles pass cyclists was found to be the most important factor in reflecting the cyclist’s experience in the city. The recorded measurements indicate that motor vehicles frequently pass at very close distances from cyclists. Since the sensor is mounted on a box approximately 8.5 cm wide attached to a bicycle with handlebars typically around 80 cm wide, measured distances below one metre correspond to situations in which vehicles pass even closer to the cyclist's body.

While DATA-RIDE cannot determine the causes of these interactions, the findings suggest that close overtaking deserves further attention from public authorities, mobility professionals, and the wider public. Citizen-generated data can help make these everyday experiences visible and provide an additional source of evidence to inform discussions on cycling conditions, infrastructure design, and coexistence between different mobility modes in Nancy and the Lorraine region.

Our perspective is that citizen-generated data can complement traditional planning tools by bringing lived experience into the conversation through a co-created dataset and helping connect mobility policies with the realities experienced by people who cycle every day.

PLUS D'ESPACE POUR LES CYCLISTES

1 mètre
minimum
de distance
de dépassement



Les données citoyennes révèlent où les rues se sentent contraignantes ou dangereuses.

DES DONNÉES POUR DES RUES PLUS SÛRES.

REJOIGNEZ LA CAMPAGNE

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DATA-RIDE

Des données citoyennes pour améliorer les conditions de pratique du vélo à Nancy.

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What We Observed

Several observations emerged consistently throughout both the measurement campaign and subsequent collective discussions.

First, potential overtaking manoeuvres were identified by combining measurements from the left-side Time-of-Flight sensor with an on-device machine-learning indicator. In total, 1,324 events were detected, of which 833 had a minimum lateral distance below one metre, representing 62.9% of all detected events. After applying the Porto precision benchmark[1] of 47.1%, this corresponds to an estimated 624 likely true overtaking events.

Some streets stood out because several of their mapped segments contained at least one event below one metre. Avenue de la Libération had 29 such segments, with an average overtaking distance of 72.5 cm. Avenue du Général Leclerc and Boulevard de l'Europe each had 23 affected segments, with average distances of 77.4 cm and 95.3 cm respectively. Rue Raymond Poincaré had 18 affected segments, with an average distance of 75.7 cm.

Second, the presence of cycling infrastructure alone does not necessarily guarantee safer interactions. Based on the cycling infrastructure dataset provided by the Métropole du Grand Nancy, 37.3% of recorded close passes below one meter occurred on streets without cycling infrastructure. However, the highest proportion of critical overtaking situations was observed on painted cycle lanes, where 65.2% of recorded overtaking events occurred below one metre.

Third, the space available to cyclists remains constrained in several locations across the city. Using the parking dataset provided by the Métropole du Grand Nancy, the analysis showed that, near on-street parking, 23.4% of the mapped corridor length had an average width below one metre. This corresponds to approximately 65 kilometres of street segments where cyclists may be more exposed to conflicts with parked vehicles, including the risk associated with opening car doors.

Taken together, these observations suggest that cycling safety cannot be understood solely through the presence or absence of cycling infrastructure. The quality of interactions among different mobility users also deserves attention, even though it shouldn't be disconnected from the quality of the current cyclist infrastructure.

Learnings During the Process

During discussions with cyclists, researchers and local actors, participants repeatedly stressed that data should not be interpreted independently from everyday cycling practices.

Several examples illustrated this point. Participants referred to cycle lanes located close to parked vehicles and within the "door zone", explaining that these

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[1] *The benchmark was conducted in Porto in March 2026 using the same senseBox:bike setup, with video footage from a 20-minute ride used to identify and classify overtaking events.*

infrastructures can sometimes feel less safe than riding in the traffic lane itself. Others noted that cycling training programmes frequently encourage cyclists to occupy the lane ("take the lane") in order to discourage dangerous overtaking manoeuvres.

These observations reinforced an important conclusion: the presence of cycling infrastructure does not automatically guarantee safer cycling conditions, it needs some improvement to be “good enough”. Complementary, the workshop discussions highlighted another important dimension: the quality and continuity of cycling infrastructure remain fundamental for bicycle adoption. Participants repeatedly stressed that the everyday use of bicycles depends less on the existence of isolated cycling facilities than on the continuity, width and overall quality of a coherent cycling network.

Several recurring issues were identified:

- discontinuous cycling routes;
- inconsistent road markings;
- narrow cycling facilities located between parked vehicles and moving traffic;
- cycling corridors where spatial constraints create uncomfortable or unsafe conditions;
- infrastructure that formally exists but does not always provide sufficient protection from close overtaking by motor vehicles.

Although participants also identified examples of infrastructure that had recently been improved and now responded more effectively to cyclists’ needs, such as on Rue Jeanne d’Arc, where the cycle lane is clearly separated from both the pavement and motor-vehicle traffic, the overall challenge remains. It is not simply about increasing the quantity of cycling infrastructure, but about developing a coherent network of “good enough” infrastructure — continuous, appropriately dimensioned and sufficiently separated from other modes to provide cyclists with comfortable and safer everyday cycling conditions.

Understanding Close Overtaking

Among recorded overtaking events, 62.9% occurred at less than one metre from the cyclist.

Potential overtaking manoeuvres were identified from short-term changes in the left-side Time-of-Flight sensor. The method did not classify every low-distance measurement as an overtaking event. Instead, it searched each trip using two-second windows and identified patterns in which the lateral distance dropped and changed rapidly, suggesting that a moving object was passing the cyclist.

Only valid distance measurements between 30 and 250 centimetres were considered. Periods in which the bicycle was likely stationary, together with a two-second buffer before and after them, were removed to reduce false detections near traffic lights, stops and intersections. Repeated patterns resembling the cyclist passing parked cars or moving through congested traffic were also filtered out.

Nearby detections were then merged into individual events. Each event was described using indicators such as minimum passing distance, distance variation, available corridor width, road roughness and the overtaking probability generated by the on-device machine-learning model. Only events with a maximum machine-learning score of at least 0.20 were retained. Scores between 0.20 and 0.70 were classified as medium confidence, while scores of 0.70 or above were classified as high confidence.

The senseBox sensor is mounted on a device approximately 8.5 centimetres wide attached to a bicycle whose handlebars are typically around 80 centimetres wide. Distances recorded below one metre therefore correspond to situations in which vehicles pass extremely close to cyclists in real-world conditions.

The parameters were based on previous testing, although future applications in Nancy would benefit from calibration using recorded rides and video validation. A more detailed description of the analysis methods is available in the document Data Processing Methodology, published on the DATA-RIDE project website in the project cloud https://cloud.reedu.de/s/JW2qiXm8nD8BLie

For many participants, this helped translate abstract numbers into lived experience.

While the project cannot determine the intention behind these manoeuvres, it does suggest that close overtaking is sufficiently common to deserve further discussion and investigation.

A Collective Hypothesis

Following the analysis of the collected data and discussions during the public workshop, participants converged around a common interpretation:

Cycling safety emerges from interactions between mobility modes.

The findings suggest that safety cannot be reduced to a simple question of infrastructure provision. Rather, cycling conditions emerge through a combination of infrastructure design and quality, available space, overtaking behaviour, interactions between cyclists, drivers and pedestrians, and shared understandings of how public space should be used.

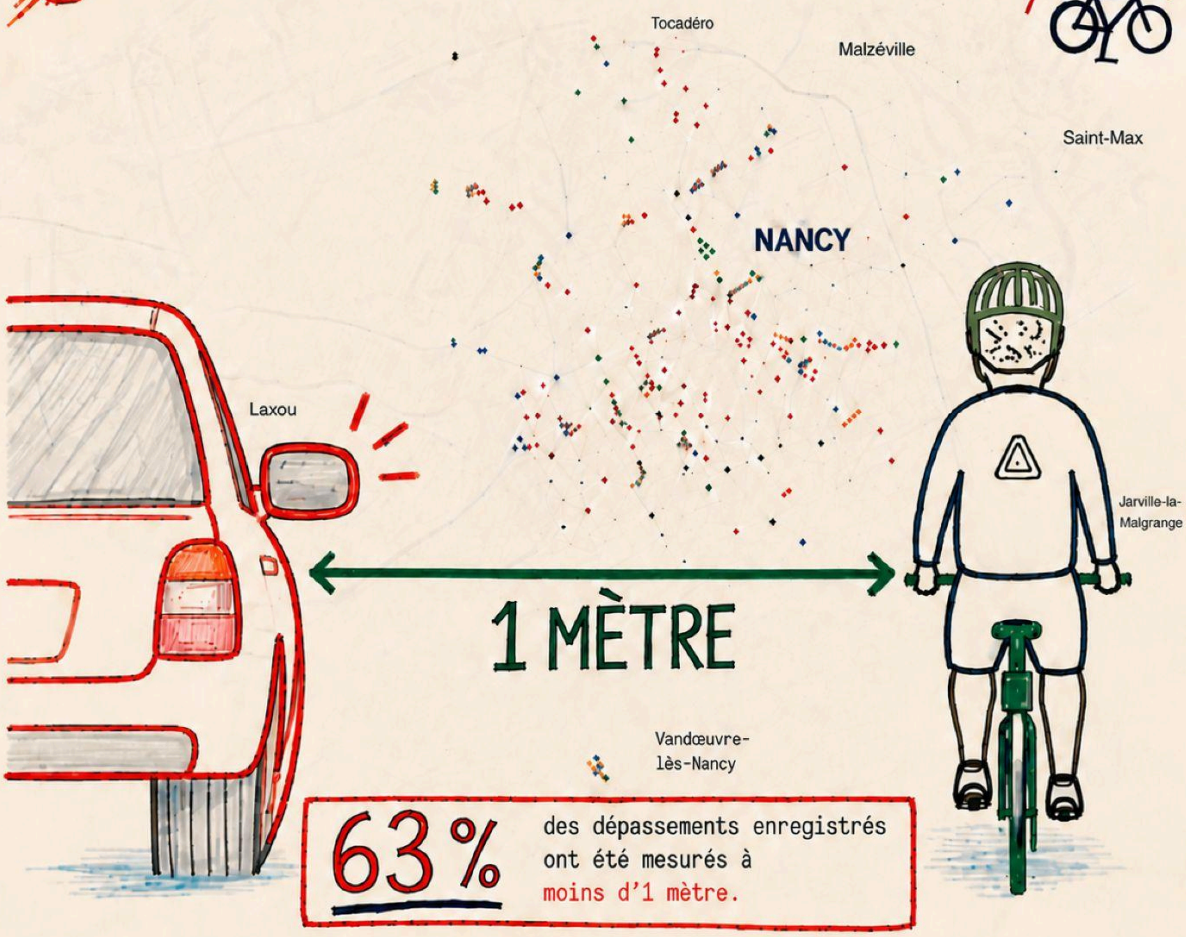
This perspective shifts attention from infrastructure alone towards the broader ecosystem of mobility behaviours and spatial practices that shape the cycling experience.

The workshop also generated complementary insights that deserve further exploration. Participants highlighted the importance of calm and cyclable neighbourhoods, including traffic-calming measures, protected cycling infrastructure, greener public spaces, and participatory planning processes. Discussions also emphasised the need to reallocate street space towards active mobility through more continuous cycling corridors, better connections between the city centre and peri-urban areas, and stronger integration between cycling and public transport systems.

Finally, participants identified night-time safety and visibility as important but often overlooked determinants of cycling adoption. Concerns included lighting gaps along major cycling routes, feelings of insecurity after dark, and opportunities to improve visibility through lighting, reflectors, and small support infrastructures. While these themes were not explored in depth through the current dataset, they emerged as relevant directions for future investigation and action.

1 MÈTRE

Le minimum n'est pas une zone de confort.



63 % des dépassements enregistrés ont été mesurés à moins d'1 mètre.

LAISSONS PLUS D'ESPACE AUX CYCLISTES.

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Communication as Part of the Solution

Participants agreed that communication campaigns should avoid blaming any particular group and instead promote mutual awareness and shared responsibility among road users.

Successful public campaigns often rely on a small number of simple and memorable messages rather than complex technical explanations. This observation directly informed the communication materials developed through the project workshop.

The campaign concept that emerged from the discussions focuses on a simple idea:

Cycling safety is a shared responsibility.

Creating safer cycling conditions requires not only infrastructure improvements, but also greater awareness of how everyday behaviours affect the experience of others in public space.

Recommendations

Based on the findings generated through DATA-RIDE and the collective discussions that followed, participants identified four complementary directions for future action.

1. Promote Safe Overtaking and Mutual Respect

Strengthen awareness regarding safe overtaking distances and respectful coexistence between mobility modes.

Possible actions include:

- public awareness campaigns (for example, targeting drivers);
- communication materials in public space;
- educational initiatives;
- integration into cycling and driving training programmes;
- promotion of existing regulations regarding overtaking distances.

The objective is not only to remind road users of legal requirements, but also to foster a culture in which adequate overtaking distance becomes visible, understandable and socially expected.

2. Support New Cyclists Through Peer Learning

Encourage programmes that help new cyclists gain confidence in everyday cycling.

Participants highlighted examples such as mentoring schemes in which experienced cyclists accompany beginners along routes considered safer and more comfortable until they feel confident enough to cycle independently (for example, Bike Anjo).

Such initiatives can help reduce perceived barriers to cycling and complement infrastructure investments.

3. Improve the Quality and Continuity of Cycling Infrastructure

Undertake a review of existing cycling routes and strategic corridors across Nancy and the wider metropolitan area.

Particular attention should be given to:

- continuity of routes;
- width and usability of cycling facilities;
- conflicts with parked vehicles;
- locations where close overtaking appears to be frequent;
- links between major destinations and everyday mobility needs.

The objective should not simply be to expand infrastructure, but to progressively achieve what participants described as a **"good enough cycling network"**: one that feels safe, coherent and attractive for everyday users.

4. Continue Developing Citizen-Generated Mobility Data

DATA-RIDE demonstrated the potential of citizen-generated data to complement traditional planning tools.

Participants therefore encourage continued collaboration between Université de Lorraine, local authorities, cycling organisations and other relevant stakeholders to expand data collection efforts and explore how such information can support mobility planning, infrastructure prioritisation and public debate.

Particular value may lie in combining citizen-generated data with existing planning datasets in order to identify priorities and monitor improvements over time.

How DATA-RIDE Looks Forward

The value of DATA-RIDE lies not only in the findings presented here but also in the process that generated them.

The project demonstrates how open technologies, citizen participation and collective interpretation can contribute to public debate on mobility and urban space.

By making everyday cycling experiences more visible, DATA-RIDE seeks to contribute to a city where cycling is not only possible, but comfortable, respected and safe.

Ultimately, DATA-RIDE should be understood as a starting point rather than a conclusion. The project opens a conversation about how citizen-generated evidence can support more informed decisions and help build cycling conditions that respond to the realities experienced by people who use the city every day.

-> Glossary

Citizen-Generated Data

Data collected, produced or interpreted by citizens rather than exclusively by public authorities, researchers or private companies. In DATA-RIDE, this refers to cycling-related data collected by volunteers during their everyday journeys.

Citizen Science

A research approach that actively involves citizens in the production of knowledge, including data collection, interpretation and discussion of results.

senseBox:bike

An open-source sensor system mounted on bicycles and connected to a smartphone. The device can collect environmental and mobility-related data, including overtaking distances between cyclists and motor vehicles.

Close Overtaking

An overtaking manoeuvre in which a motor vehicle passes a cyclist at a very short distance. Within the DATA-RIDE project, particular attention was given to overtaking events recorded at less than one metre from the bicycle.

Cycling Infrastructure

Physical facilities designed to support cycling, including cycle lanes, protected cycle tracks, shared-use paths, cycling corridors, crossings and bicycle parking facilities.

Painted Cycle Lane

A cycling lane separated from motor vehicle traffic only through road markings and without physical protection.

Door Zone

The area adjacent to parked vehicles where opening a car door may create a collision risk for cyclists.

Traffic Calming

Urban design and traffic management measures intended to reduce vehicle speeds and improve safety, such as 30 km/h zones, one-way systems, modal filters and street redesign.

Active Mobility

Modes of transport based primarily on human physical activity, including walking, cycling and the use of other non-motorised mobility devices.

Modal Filter

A street design intervention that restricts through-traffic by motor vehicles while maintaining access for cyclists and pedestrians.

Good Enough Cycling Network

A term emerging from the DATA-RIDE discussions referring to a cycling network that is sufficiently continuous, coherent, comfortable and safe to support everyday bicycle use by a broad range of people.

Cycling Mentoring

Programmes in which experienced cyclists accompany less experienced riders to help them gain confidence, identify safer routes and develop everyday cycling skills.

Coexistence Between Mobility Modes

The ability of different users of public space—cyclists, pedestrians, drivers and public transport users—to move safely and respectfully within the same urban environment.

Open-Source Technology

Technology whose design and specifications are publicly available and can be freely used, adapted and improved by others.

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*illustrations based on visuals co-created collectively throughout the DATA-RIDE development process.